

Ashwin Badamikar

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EDUCATION

Northeastern University

Master of Science in Information Systems – GPA: 3.5/4.0

Boston, MA

Expected Dec 2026

JSS Academy of Technical Education

Bachelor of Engineering in Computer Science and Engineering

Bengaluru, India

2019 – 2023

TECHNICAL SKILLS

AI/ML Frameworks: LangGraph, LangChain, OpenAI API, Mistral AI, Ollama, Hugging Face Transformers, TensorFlow, PyTorch
RAG & Vector Search: Pinecone, ChromaDB, pgvector (PostgreSQL), Retrieval-Augmented Generation, Embedding Models
Languages & Libraries: Python (Pandas, NumPy, PySpark), SQL, JavaScript, Bash, NLP, Deep Learning, Statistical Modeling
Data & Cloud: Snowflake, Supabase, Apache Airflow, Databricks, AWS, Azure ML, Docker, GitHub Actions, FastAPI, Streamlit
Concepts: Multi-Agent Systems, Prompt Engineering, LLM Observability, MLOps (MLflow), Generative AI, Fine-Tuning
Certifications: What Is Generative AI? (LinkedIn) Quantum-Safe Cryptography (IBM) Cloud Computing Architecture (Great Learning)

WORK EXPERIENCE

AI Developer Intern

Jan 2026 – June 2026

Aosenuma Inc.

Remote, USA

- Developed **multi-agent AI workflows** using LangGraph to automate ESG research and sustainability scoring on the StakeAI platform, orchestrating specialized agents for data collection, semantic analysis, and structured report generation.
- Built **RAG pipelines** integrating Pinecone vector search with FastAPI microservices and Supabase, enabling context-aware AI responses grounded in large ESG document corpora with sub-200ms retrieval latency.
- Implemented **Helicone LLM observability** to monitor token usage, latency, and error rates across all production AI pipelines; applied prompt engineering strategies that reduced hallucination rates by **22%** across evaluated outputs.

Software Engineer

Jan 2023 – Jul 2024

TurnOn

Bengaluru, India

- Developed **Python-based data pipelines and automation systems** forming the feature engineering backbone for ML model training workflows, supporting AI-powered automation in the agriculture and construction industries.
- Built **real-time event streaming infrastructure** using Apache Kafka to process 100K+ daily events, enabling downstream AI systems to consume live operational data for predictive modeling and anomaly detection.
- Collaborated with data science teams to design **feature stores and preprocessing pipelines**, improving model training data quality and reducing feature engineering lead time by **30%**.

ACADEMIC PROJECTS

Chronic Care AI – Multi-Agent Healthcare Assistant

- Architected a **HIPAA-compliant multi-agent system** using LangGraph with specialized agents for symptom analysis, medication lookup, and care coordination; integrated Google Maps and Edamam APIs for geolocation-aware care recommendations.
- Implemented **Mistral embeddings with Pinecone** for semantic retrieval over clinical notes, achieving a **28% improvement** in retrieval precision; deployed end-to-end via Docker and GitHub Actions CI/CD on a FastAPI and Snowflake backend.

NBA RAG System – Conversational Sports Intelligence

- Built an **end-to-end RAG pipeline** using PostgreSQL with pgvector for dense embedding storage and ANN similarity search, with fully offline local LLM inference via Ollama (llama3.2:3b and nomic-embed-text) for privacy-preserving responses.
- Designed an **Angular chat UI** with real-time streaming and a FastAPI backend, enabling natural language Q&A over NBA player statistics, team performance data, and game records with sub-second latency.

Nutrition Tutor Bot – Multimodal RAG Chatbot

- Built a **multimodal RAG chatbot** combining GPT-4 Vision for food image analysis with ChromaDB for semantic retrieval; integrated USDA Nutrition API as a live data source to keep dietary information current and grounded.
- Designed a **hybrid retrieval strategy** blending dense semantic search and sparse keyword matching; benchmarked chunking strategies and embedding models to maximize answer relevance and end-to-end response accuracy.

Diabetes Treatment Recommendation System – Reinforcement Learning

- Developed a **reinforcement learning agent** for personalized diabetes treatment recommendations on Azure ML, with custom reward shaping to jointly optimize for glycemic control and long-term patient treatment adherence.

RESEARCH & PUBLICATIONS

Research Assistant – NLP & Machine Learning

2022 – 2023

JSS Academy of Technical Education

Bengaluru, India

- **Published research** on cross-lingual transfer learning for NLI using the XNLI benchmark across 15 languages; findings **awarded by the Karnataka State Council for Science and Technology (KSCST) in association with IISc, Bengaluru.**
- Designed **end-to-end experimental pipelines** for data preprocessing, model training, hyperparameter tuning, and statistical analysis; produced formal peer-reviewed research findings with rigorous documentation.